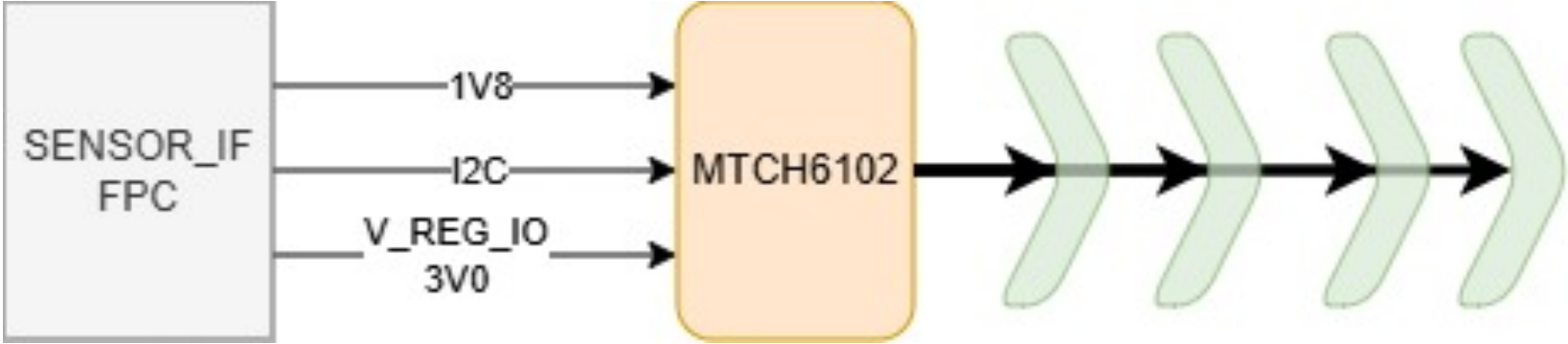


Schematic Sheets

#	Name	File	Description
1	Main	SensWear_Touch	Touch Controller IC and Electrodes
2			
3			
4			

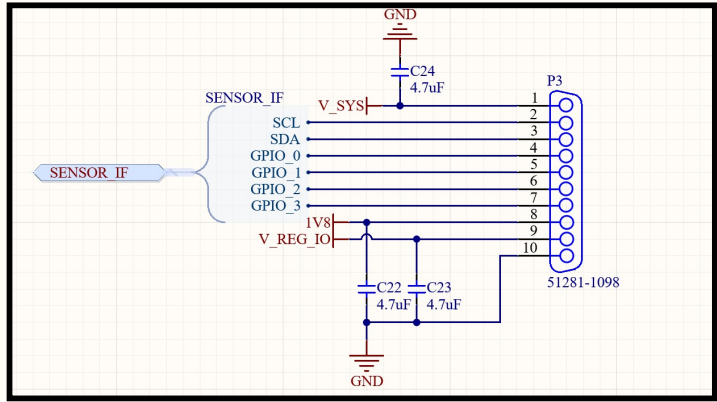
Revision History

Revision	Date	Description
A0	May 2025	Initial validation design on rigid PCB
B0	March 2026	Flexible PCB manufacturing updates



SensWear-V1R1

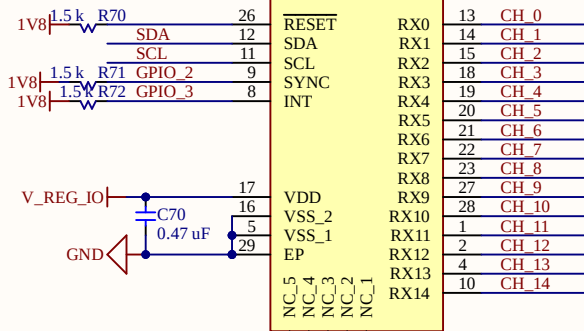
Designer/Engineer	Board Title		
Yusein Ali	SensWear-Touch-RevA		
Drawn By	Document Name		
Yusein Ali	SensWear-V1R1-Touch-		
Approved By	Ver.	Rev.	Date
Dariush Salami	1	B0	01.03.2026



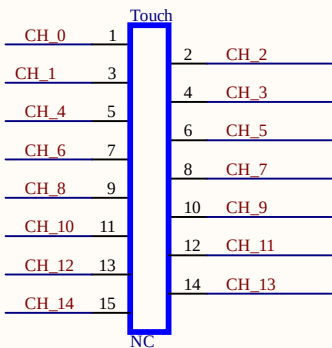
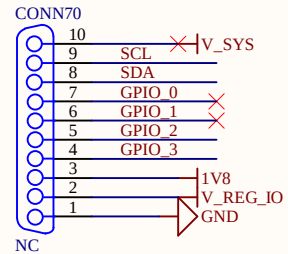
MTCH6102 is not recommended for new designs
Use 3V for VDD

I2C Address: 0100 101X

U70



MTCH6102-I/MV



TOUCH SENSOR (MTCH6102) USAGE NOTES
Supply: V_REG_IO / VDD (1.8V–3.6V)
Note : Not recommended for new designs

[1. INTERFACE]

Protocol : I2C (I2C21)
Speed : Up to 400 kHz
Address : 0x25, 7-bit Zephyr address
0x4A write / 0x4B read
0100 101X

[2. PIN MAPPING]

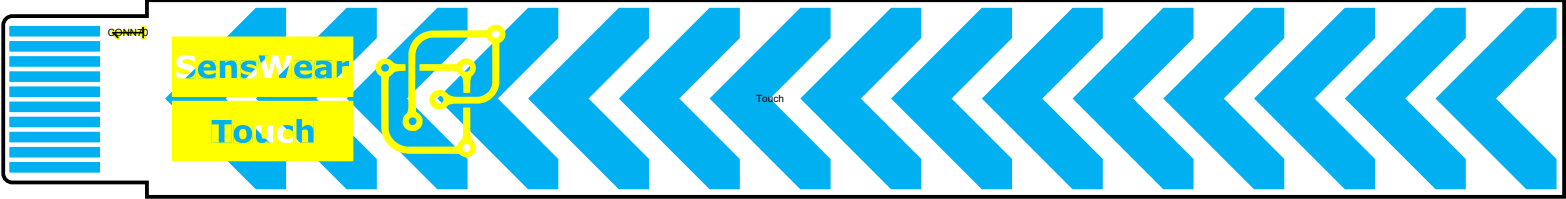
Signal	MTCH6102 Pin	Net
SCL	Pin 11	SYS_I2C_SCL (P1.03)
SDA	Pin 12	SYS_I2C_SDA (P1.02)
RESET#	Pin 26	RESET
SYNC	Pin 9	GPIO_2 (P1.12)
INT#	Pin 8	GPIO_3 (P1.11)
VDD	Pin 17	V_REG_IO
VSS	Pins 5,16	GND
RX0-RX14	Pins 13...28	CH0-CH14

[3. NOTES]

INT# is active-low open-collector.
SYNC is active-high frame status output.
RESET has internal pull-up

		SensWear-V1R1	
Designer/Engineer Yusein Ali		Board Title SensWear-Touch-RevA	
Drawn By Yusein Ali		Document Name SensWear_Touch.SchDoc	
Approved By Dariush Salami		Rev. B0	Sheet 1 of 1 Date 01.03.2026 Size A4

View from Top side (Scale 4:1)



View from Bottom side (Scale 4:1)



Bill Of Materials

Line #	Designator	Comment	Quantity
1	C70	0.47 uF	1
2	CONN70	NC	1
3	R70, R71, R72	1.5 k	3
4	Touch	NC	1
5	U70	MTCH6102-I/MV	1